AI for Medicine – Project Report

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1. Project Title

Diabetic Retinopathy Severity Grading: A Transfer-Learning Approach on the APTOS 2019 Fundus Image Dataset

2. Problem Statement

Diabetic retinopathy (DR) is a microvascular complication of diabetes mellitus and one of the leading causes of preventable blindness in working-age adults worldwide [1]. Early detection enables sight-saving interventions, but screening still relies on direct examination of fundus photographs by ophthalmologists — a process that is time-consuming, requires specialised training, and is unevenly distributed across geographies. Deep-learning-based image grading has emerged as a complement to expert review: Gulshan et al. [2] showed that a convolutional neural network trained on more than 100,000 fundus images can match board-certified ophthalmologists on referable DR detection, and subsequent work [3] refined the preprocessing pipeline and architectures on which the present project builds.

3. Objective of the Study

The objective is to design, train, and evaluate a deep-learning classifier that, given a colour fundus photograph, predicts the severity stage of diabetic retinopathy on the five-level International Clinical Diabetic Retinopathy Disease Severity Scale [4]: No DR (0), Mild non-proliferative (1), Moderate non-proliferative (2), Severe non-proliferative (3), Proliferative (4). Three quantifiable sub-objectives are pursued. First, build a transfer-learning EfficientNet-B0 baseline on APTOS 2019 and estimate its held-out Quadratic Weighted Kappa (QWK). Second, quantify the stability of this estimate across five stratified cross-validation folds with independent seeds, reporting mean ± standard deviation under simultaneous perturbation of the data partition and the network initialisation. Third, interpret the model with Grad-CAM to confirm that predictions are driven by clinically plausible regions of the retina such as microaneurysms, exudates, and neovascularisations.

4. Dataset Description

Experiments are conducted on the APTOS 2019 Blindness Detection dataset [5], released by the Asia Pacific Tele-Ophthalmology Society on Kaggle in collaboration with the Aravind Eye Hospital (Madurai, India). The dataset consists of 3,662 colour fundus images acquired in real clinical conditions across multiple sites, with heterogeneous capture devices and image resolutions from 433×289 to 3,388×2,588 pixels. Each image is annotated with an integer severity label between 0 and 4 by trained ophthalmologists, following the International Clinical Diabetic Retinopathy Disease Severity Scale [4]. The class distribution is markedly imbalanced (Table 1).

Severity class	**Label**	**Approx. prevalence**
No DR	0	≈ 49%
Mild	1	≈ 10%
Moderate	2	≈ 27%
Severe	3	≈ 5%
Proliferative DR	4	≈ 9%

Table 1. Class distribution of the APTOS 2019 training set.

Although APTOS also distributes 1,928 unlabelled images intended as the official Kaggle test set, those labels are held private by the organisers and accessible only through the submission system; following standard post-competition academic practice, we work exclusively with the 3,662 labelled images and partition them via stratified split into a 70/15/15 train/validation/test scheme. Several known issues affect the dataset: resolution variability across cameras (addressed by uniform resize and intensity harmonisation in §5), severe class imbalance (addressed by a class-balanced sampler at training time), label noise from inter-rater disagreement ($\kappa \approx 0.50$–0.65 among expert ophthalmologists [2]) and an additional dataset-specific noise documented in §6, and the absence of patient identifiers, which prevents a strict patient-level split (discussed in §6 and §9).

5. Data Preprocessing

Preprocessing follows the pipeline introduced by Ben Graham [3], winner of the 2015 Kaggle Diabetic Retinopathy competition, and adopted by the majority of top APTOS 2019 submissions. Three deterministic steps are applied identically to every image: (i) a circle crop that thresholds the grayscale image, fits the largest external contour with a minimum enclosing circle, and crops to its bounding box, removing device-dependent dark borders; (ii) bilinear resize to 224×224, the native input resolution of EfficientNet-B0; (iii) local-contrast normalisation as $I' = 4 \cdot I - 4 \cdot G(I) + 128$ with σ

= 10 Gaussian kernel, which subtracts the slowly-varying illumination and emphasises microaneurysms, haemorrhages and exudates. On-the-fly augmentation (random flips, ±15° rotations, mild colour jitter) is applied only to the training fold. The deterministic preprocessing is computed once and cached, so only the random augmentation runs during training.

The local-contrast normalisation acts as an intensity-based data harmonisation step [6]: by high-pass filtering out the device-dependent illumination, it prevents the network from learning camera-specific shortcuts (Topcon vs. Zeiss colour temperature, for instance) instead of clinically meaningful lesion patterns. This is conceptually analogous to the harmonisation pipelines developed for multi-centre MRI studies [6], applied in the image domain rather than the feature domain.

Aggressive downsampling carries the risk of erasing class-discriminative information. To rule this out, we test empirically whether the original image dimensions are correlated with the DR severity class via a Kruskal–Wallis non-parametric one-way ANOVA on per-image height, width, and total pixel area, grouped by class (Table 2).

Dimension	**H statistic**	**p-value**	**Verdict (α=0.05)**	**Range of medians**
Height	1254.72	< 0.0001	Reject null	1,050 → 1,958 px
Width	1499.60	< 0.0001	Reject null	1,050 → 2,588 px
Area	1472.58	< 0.0001	Reject null	1.10 → 5.07 Mpx^2

Table 2. Kruskal–Wallis test of class–dimension dependence on raw images.

All three tests reject the null hypothesis of equal distributions with $p < 10^{-4}$. The No-DR class images are systematically smaller and square (median 1,050×1,050 px, area ≈ 1.10 Mpx^2) while the diseased classes are larger and rectangular (median ≈ 1,958×2,588 px, area ≈ 5.07 Mpx^2). This is a structural confounder of the dataset itself: original image dimensions are nearly perfect predictors of the No-DR-vs-DR binary distinction, presumably because the No-DR images underwent a different acquisition or pre-cropping pipeline. The finding confirms — rather than undermines — the choice of uniform resize and intensity harmonisation, which actively neutralise this class–dimension shortcut and force the network to learn from the visual content of the fundus.

6. Avoiding Data Leakage

Data leakage is actively prevented by design and explicitly verified by automated assertions in the notebook. The 3,662 labelled images are partitioned with a stratified 70/15/15 split at the dataframe level with a fixed seed; a programmatic assertion confirms that the three id_code sets are pairwise disjoint. To go beyond sample-level disjointness, an MD5 hash of the full content of every image is used to detect byte-exact duplicates: the audit revealed 128 duplicate pairs across the 3,534 unique images (3.5% of the dataset). Of these, 64 are within-fold (same image, same fold —

innocuous over-weighting) and 64 are cross-fold, i.e., the same image present in two different folds. The cross-fold pairs are distributed as shown in Table 3.

Cross-fold pair	Count	Severity
train ↔ test	29	Severe — inflates test_QWK
train ↔ val	28	Mild — inflates the early-stopping target
val ↔ test	7	Severe — small but real leakage on evaluation
Total cross-fold	64	Actively removed before training

Table 3. Cross-fold byte-duplicate inventory and impact.

All 64 cross-fold duplicates were eliminated following a priority that protects the evaluation set: when a duplicate spans test and another fold, the copy outside test is removed; when it spans train and val, the train copy is removed. Concretely, 56 ids were removed from train and 7 from val; the test fold remained intact at 550 images. Post-de-duplication, the disjointness assertion still passes and the fold sizes become train = 2,506, val = 543, test = 550.

Two residual sources of leakage are explicitly acknowledged. First, 30 of the 128 duplicate pairs are byte-identical images carrying conflicting severity labels — same fundus, different graders, different annotations. This intra-dataset annotation inconsistency is independent of any split choice and places an irreducible upper bound on the QWK achievable by any classifier. Second, APTOS 2019 does not distribute patient identifiers: both eyes of the same subject can in principle appear across train, validation and test folds, producing a small residual leakage at the patient level that cannot be removed without external metadata. Both effects bias the absolute QWK upwards by an amount that cannot be quantified with the data at hand.

7. Machine Learning Pipeline

The classifier is built on EfficientNet-B0, a compound-scaled convolutional network that offers a favourable accuracy/compute trade-off and is well suited to a Colab T4 GPU runtime. The backbone is initialised with ImageNet-pretrained weights via the timm library; the original 1,000-way classifier head is replaced with a single linear layer mapping the global-pool features to five logits. Transfer learning is essential at this dataset scale: random initialisation would overfit catastrophically given the ratio of ~5M parameters to ~2.5k training samples.

Training uses AdamW (learning rate $3{\cdot}10^{-4}$, weight decay 10^{-4}), cosine annealing over 15 epochs, mixed-precision FP16 for ~30% speed-up, and early stopping with patience of 4 epochs on the validation QWK. Class imbalance is addressed at training time by a WeightedRandomSampler with inverse-frequency weights. To quantify the stability of the result, the experiment is repeated under stratified 5-fold cross-validation on the de-duplicated train ∪ val pool (3,049 images); the test fold remains globally held out. K = 5 is a standard methodological choice for medical-imaging datasets of order 10^3–10^4 samples and is not a hyperparameter to be tuned. Each fold uses a distinct seed (SEED + 100·k) that simultaneously perturbs the partition, the random initialisation of the new classification head, the shuffle order, and the augmentation stream, satisfying the two methodological requirements of split-seed and weight-init variation with a single parameter.

The primary evaluation metric is the Quadratic Weighted Kappa (QWK), the official metric of the APTOS 2019 competition. QWK is a Cohen's kappa weighted by the squared distance between predicted and true classes on the ordinal severity scale, so errors close to the true class incur small penalty and errors far from the true class incur large penalty — matching the ordinal nature of DR grading. Secondary metrics include overall accuracy, macro F1, per-class precision/recall, and the row-normalised confusion matrix. All multi-class predictions are taken as the argmax of the post-softmax probability vector; no per-class operating threshold is tuned, consistent with the standard treatment of QWK as a ranking-aware accuracy metric.

8. Results

The first sub-objective is the baseline performance estimate. The representative single-run model, trained with seed = 42 on the de-duplicated 70/15/15 partition, achieves on the held-out test fold a QWK of 0.821, accuracy of 0.755, and macro-F1 of 0.553. The per-class report (Table 5) shows the structural imbalance that drives the aggregate macro-F1: near-perfect performance on the No-DR class (F1 = 0.961) and progressively weaker performance on the rarer, clinically more severe categories. Notably, the per-class one-vs-rest AUC values (Figure 2) all exceed 0.88, indicating that the model has good discriminative power per class; the lower F1 scores on rare classes derive from the argmax decision rule rather than insufficient learned representation, suggesting that recalibrated per-class thresholds or a cost-sensitive loss could substantially improve recall on Severe and Proliferative.

Class	Precision	Recall	F1-score	Support
No DR	0.970	0.952	0.961	271
Mild	0.393	0.393	0.393	56
Moderate	0.659	0.720	0.688	150
Severe	0.353	0.207	0.261	29
Proliferative	0.447	0.477	0.462	44
macro avg	**0.564**	**0.550**	**0.553**	**550**
weighted avg	**0.752**	**0.755**	**0.752**	**550**

Table 5. Per-class metrics of the representative run on the 550-image test fold.

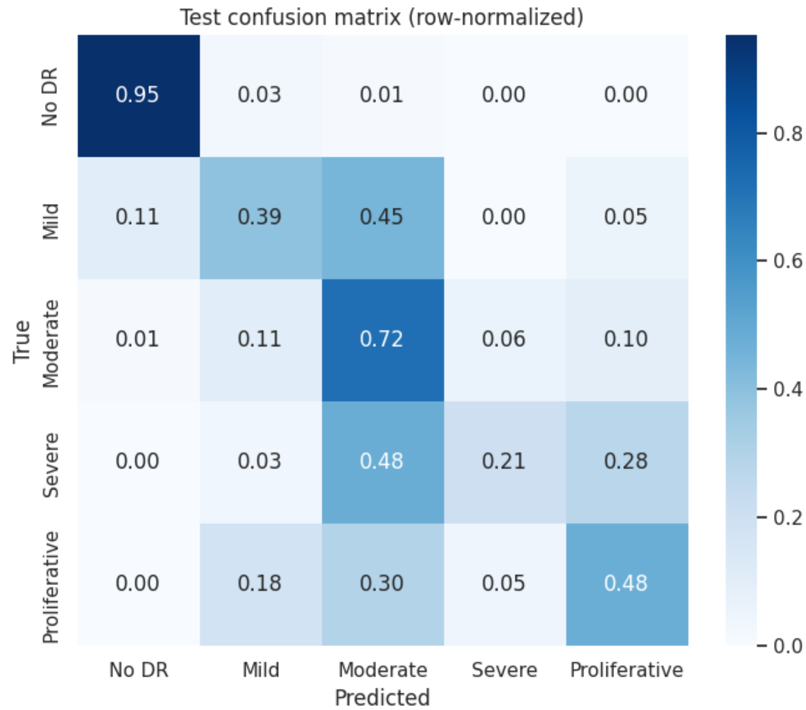

Figure 1. Row-normalised confusion matrix on the held-out test fold (representative run, seed = 42). Each row sums to 1 and shows the distribution of predicted classes for samples with a given true label.

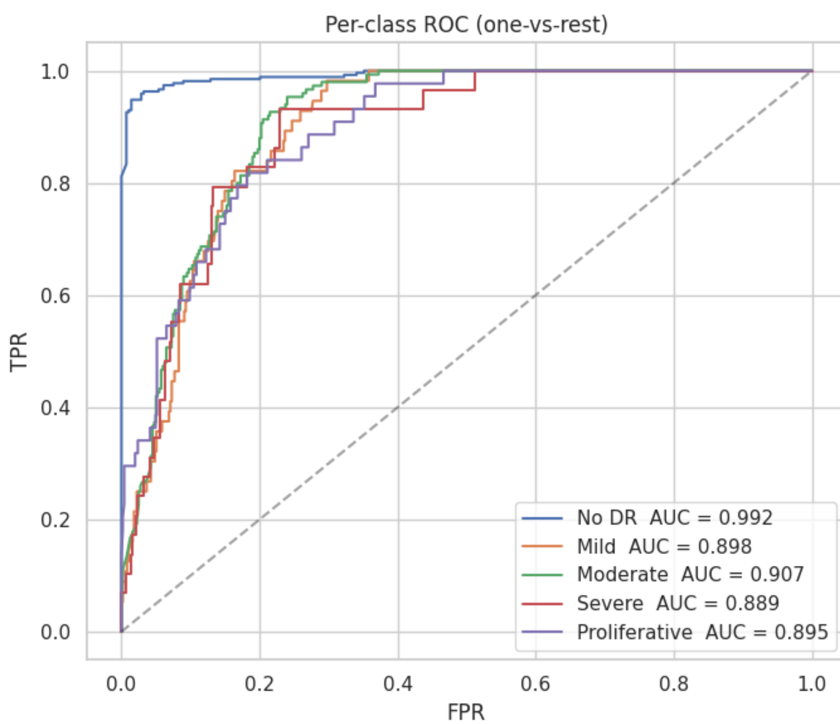

Figure 2. One-vs-rest ROC curves on the test fold, with per-class AUC values.

The second sub-objective is the stability of this estimate under simultaneous perturbation of data partition and network initialisation. Quantified by 5-fold cross-validation with independent seeds, the test QWK averages 0.844 ± 0.008, with a coefficient of variation of 0.95% across folds (Table 4). The Shapiro–Wilk test on the five fold-level QWK values does not reject normality ($p = 0.13$), so the parametric mean ± std is reported as the primary summary; the non-parametric median [IQR] is reported alongside for transparency. The representative run lies within ≈ 3 σ of the CV mean, on the lower tail of the fold distribution. For external comparison, published single-model baselines on APTOS 2019 typically report QWK in the 0.85–0.91 range on internal test folds; the top of the Kaggle leaderboard (attainable only with multi-model ensembles and TTA) reaches 0.93–0.94 on the held-out competition test set, which is not directly comparable with our internal fold.

Estimator	**Accuracy**	**Macro F1**	**QWK**
Representative run (seed=42)	0.755	0.553	0.821
5-fold CV (mean ± std)	0.788 ± 0.007	0.605 ± 0.023	0.844 ± 0.008
5-fold CV (median [IQR])	0.786 [0.78,0.79]	0.609 [0.58,0.62]	0.847 [0.84,0.85]

Table 4. Aggregate performance: representative run vs. 5-fold cross-validation.

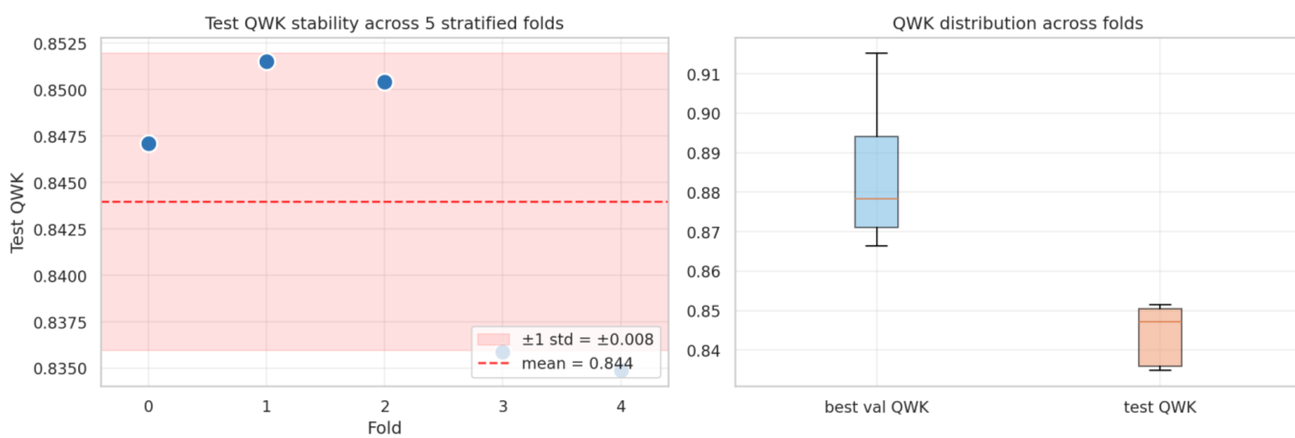

Figure 3. Test QWK across the five stratified cross-validation folds. Left: per-fold trajectory with mean and ±1 std band. Right: boxplot comparison of validation and test QWK distributions.

The third sub-objective is the qualitative validation of the learned representation through Grad-CAM. The Grad-CAM panel (Figure 4) confirms that the model's predictions are driven by clinically plausible regions (vascular abnormalities, exudates, haemorrhages), supporting the validity of the learned representation beyond the raw accuracy figures.

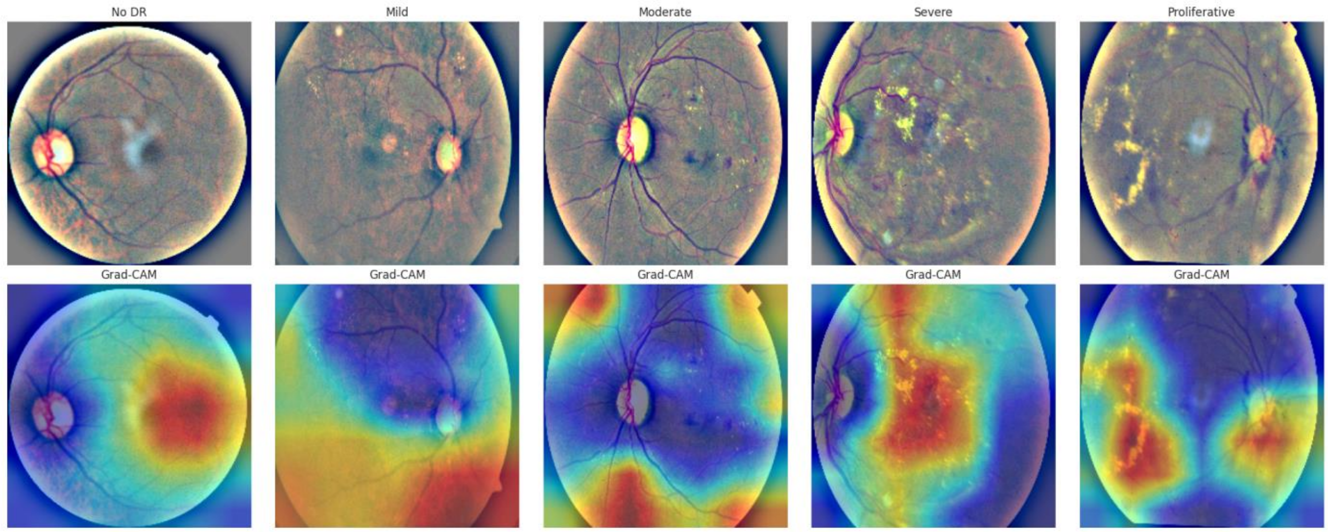

Figure 4. Grad-CAM heatmaps overlaid on representative test images, one per severity class. Warmer colours indicate spatial regions of higher attribution. The highlighted areas correspond to clinically meaningful lesions (microaneurysms, exudates, neovascularisations) in the diseased classes.

9. Discussion

The de-duplicated pipeline reaches a cross-validated test QWK of 0.844 ± 0.008, comparable to single-model published baselines on APTOS 2019 and well above the inter-rater agreement of expert ophthalmologists ($\kappa \approx 0.50$–0.65). The performance is highly non-uniform across classes: No DR is classified almost perfectly (F1 = 0.961), while Severe (F1 = 0.261) and Proliferative (F1 = 0.462) are substantially weaker. From a clinical standpoint these are the most consequential limitations: false negatives on advanced DR correspond to patients who should be urgently referred for sight-saving therapy and who, without referral, are at high risk of irreversible visual loss. The coefficient of variation of just under 1% on test QWK indicates that the pipeline is stable under seed perturbation; the larger variability in macro F1 (~4%) is concentrated in the rare classes, consistent with their small support.

An instructive observation from the de-duplication audit is that removing the cross-fold byte-duplicates reduced the representative-run test QWK from 0.854 to 0.821 and disproportionately depressed the Severe F1 (0.49 → 0.26). This indicates that a non-trivial fraction of the apparent performance on rare classes was attributable to inadvertent leakage, where the same image appearing in both training and test artificially boosted recall on the categories where any signal is scarce. The cleaner pipeline gives a more honest estimate and exposes that the network's capacity to learn the rarest classes from such small support is genuinely limited.

Four limitations bound the interpretation of the results. The experiments use a single-source dataset and the test fold shares its acquisition setting with the training data; performance under domain shift to other populations or cameras is not assessed and is expected to be lower, with external validation on IDRiD, MESSIDOR-2 or EyePACS left as future work. APTOS 2019 distributes no patient identifiers, so subject-level splitting — the correct practice advocated by Yagis, Diciotti et al. [7] for medical imaging — cannot be implemented. The MD5-based audit removes byte-exact duplicates, but a residual non-independence leakage (L3.2 in the taxonomy of Kapoor and Narayanan [8]) at the patient level — for example two distinct photographs of the same eye or the fellow eye of the same subject — remains unavoidable. This is a structural limitation of the dataset rather than of the pipeline. The original image dimensions are strongly correlated with the No-DR-vs-DR distinction, a structural confounder neutralised by the preprocessing but indicative of an underlying acquisition bias. Finally, 30 byte-identical image pairs carry conflicting clinical labels, an intra-dataset annotation inconsistency that places an irreducible upper bound on the achievable QWK.

10. Conclusions and Future Work

A transfer-learning EfficientNet-B0 pipeline with Ben Graham intensity harmonisation reaches a held-out test QWK of 0.844 ± 0.008 on de-duplicated APTOS 2019, with cross-validation stability (CV < 1%) that quantifies and corroborates the point estimate. The model is effective at distinguishing healthy from diseased fundi and most error-prone on the rarest, clinically most severe classes. The pipeline is end-to-end reproducible from a single notebook with a fixed seed and Kaggle-API data retrieval. Three directions warrant explicit follow-up: external validation on independent datasets to characterise domain-shift behaviour; replacement of cross-entropy with an ordinal-aware loss (CORN, CORAL) or class-balanced focal loss to address the recall asymmetry on severe cases; and self-supervised pretraining on unlabelled fundus images to reduce dependence on ImageNet features whose alignment with retinal pathology is incidental.

11. Ethics and Data Privacy

The APTOS 2019 dataset is publicly released by the Asia Pacific Tele-Ophthalmology Society on Kaggle, where competitors accept a data-use agreement that restricts the data to non-commercial research. All images are de-identified at source; no demographic information, patient identifier, or acquisition date that could allow re-identification accompanies the fundus photographs. No new data are collected for this project, and the dataset itself is not redistributed through the project repository: it is fetched at runtime via the Kaggle API and remains subject to its original terms. Third-party code dependencies (PyTorch, torchvision, timm, grad-cam) are released under permissive open-source licenses (BSD and Apache 2.0); project code is released under the MIT License.

Three considerations deserve attention beyond licence compliance. The model could serve as a screening aid but is not a diagnostic device; deployment in a clinical setting would require regulatory clearance (CE/MDR in the European Union, FDA clearance in the United States) and prospective validation. The training population (single hospital, predominantly South-Asian patients) is narrow, and the model's behaviour on other populations is uncertain; deployment without stratified per-population performance reporting would risk amplifying healthcare disparities. Finally, given the low recall on advanced DR observed in §8, automated screening must be paired with — not replace — human clinical judgement.

12. Code and Reproducibility

The complete pipeline is implemented in a single Jupyter notebook, aptos-dr-grading.ipynb, published with an embedded "Open in Colab" badge at https://github.com/JonnyCerullo/aptos-dr-grading and designed to run on Google Colab with a GPU runtime. Dependencies are PyTorch ≥ 2.0, timm ≥ 1.0, scikit-learn ≥ 1.5, scipy, opencv-python-headless, kagglehub ≥ 0.4, and grad-cam. Reproducibility is built into the pipeline from the outset: a single SEED constant seeds Python's random, NumPy, PyTorch (CPU and CUDA), cuDNN deterministic flags, and the PYTHONHASHSEED environment variable, so every random operation (split, shuffle, weight initialisation, augmentation, balanced sampling) reads from the same source. For cross-validation, each fold has a derived seed (SEED + 100·k), which makes the entire 5-fold experiment reproducible as a function of a single base seed. To reproduce: generate a Kaggle API token (KGAT_ format), store it as a Colab secret named KAGGLE_API_TOKEN with notebook access enabled, and run the notebook top to bottom; the data is fetched automatically, the Ben Graham preprocessing is cached locally on first run, and the entire experiment takes ≈ 30 min on a Colab T4 GPU.

13. References

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